Introduction to Composites Material

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Lecture 2
Fibers

On the basis of geometry of the reinforcement, they are classified as

(a) Fibrous

(b) Particulate

(c) Flakes composites

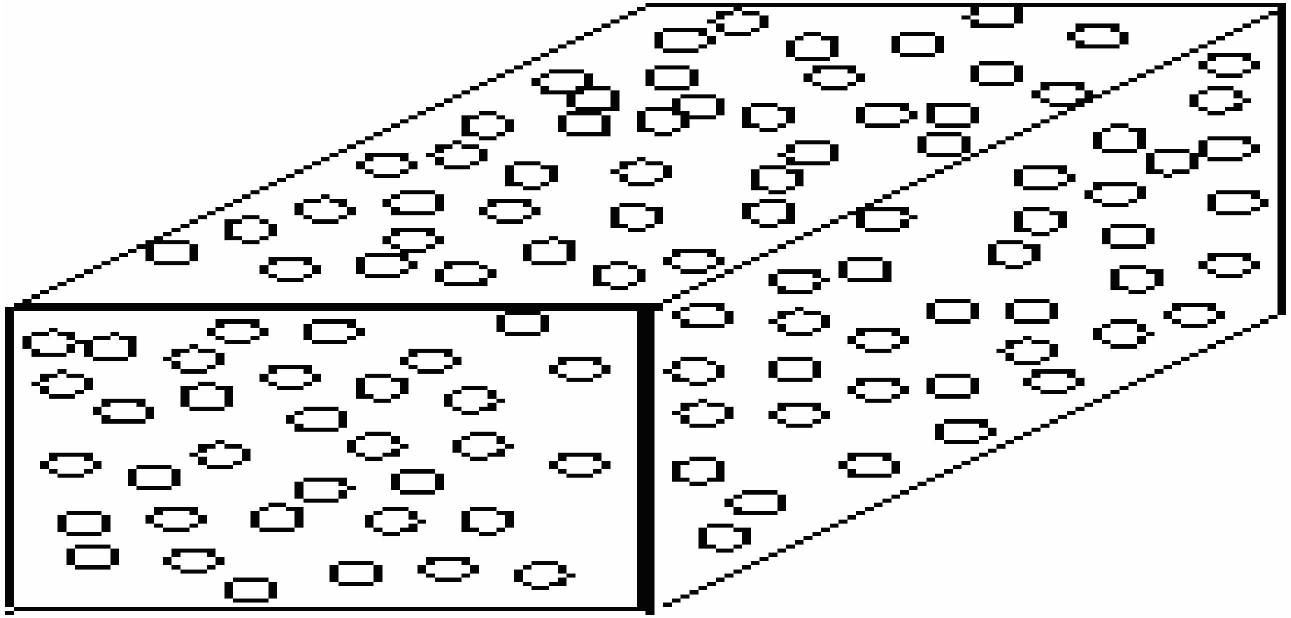
Reinforcements : Particles, whiskers, ribbons, discontinuous fibres, Continuous fibre, flakes, sheets, wire, CNT

Ribbons : rapid solidifications 20 micron

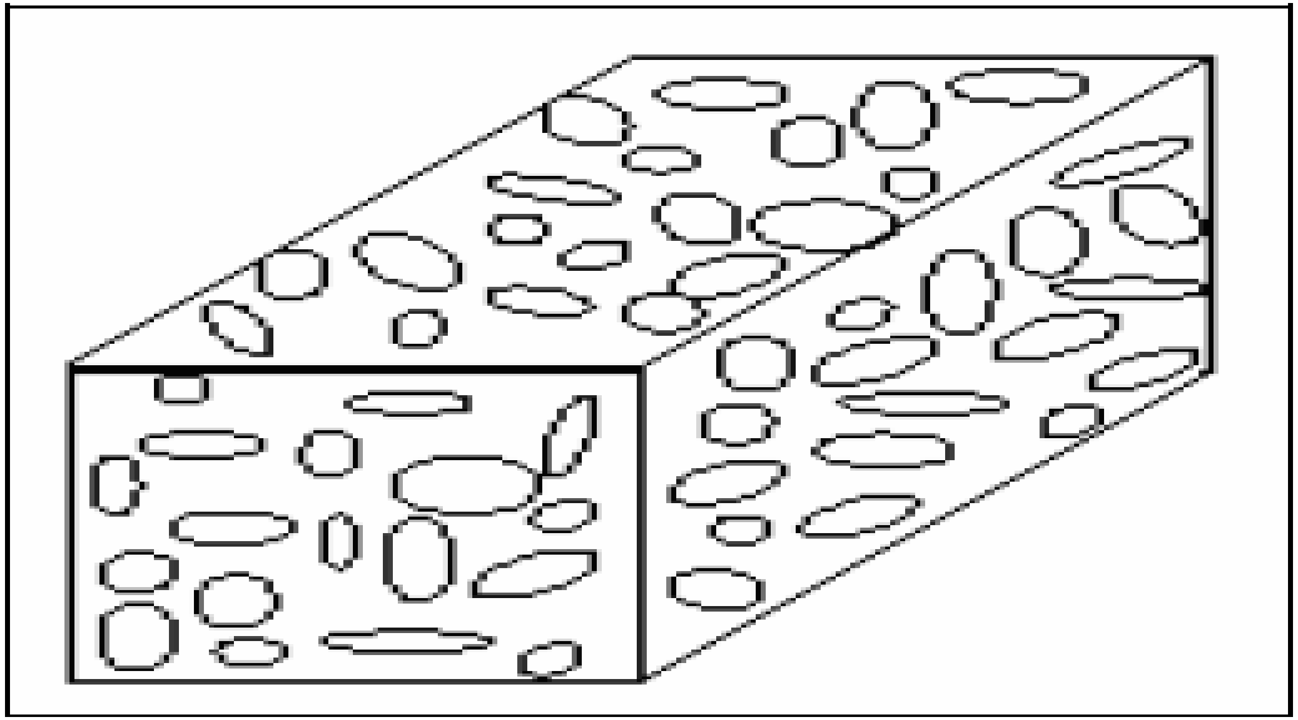
Whiskers : single crystals

Continuous fibers are widely used for increase strength and stiffness

Particle as the reinforcement (Particulate composites)



Fillers as the reinforcement (Filler composites)

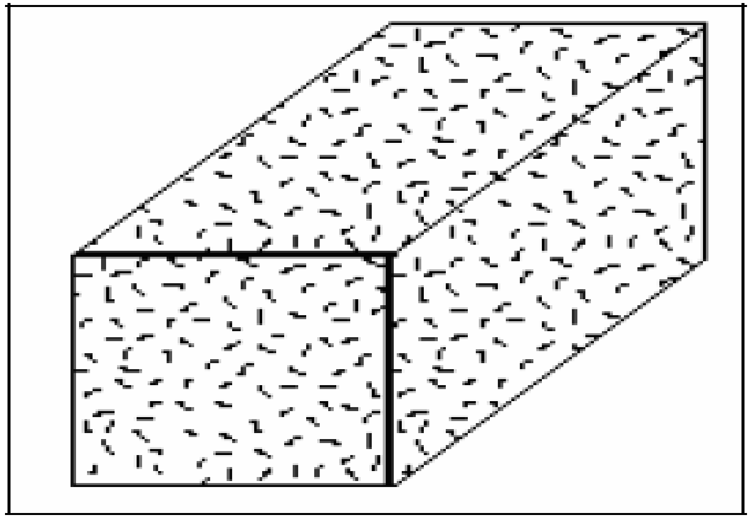


Fibre Reinforced Composites(FRP)

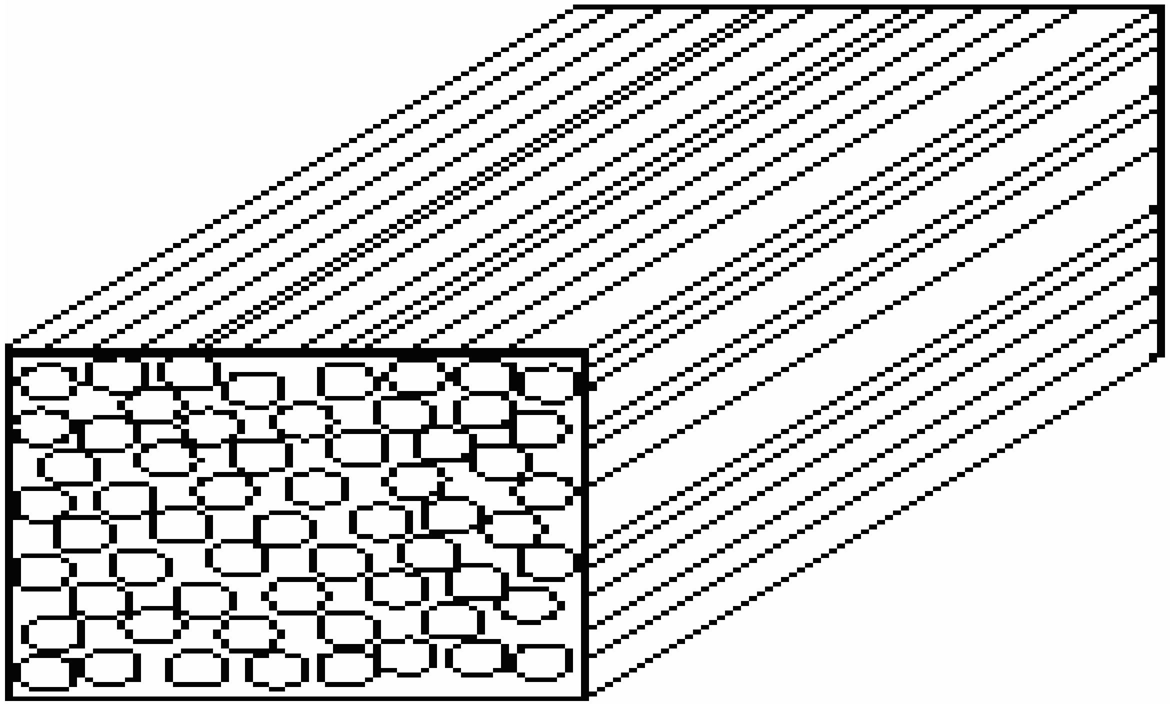
1. Continuous fibre reinforced composites

2. Discontinuous fibre reinforced composites

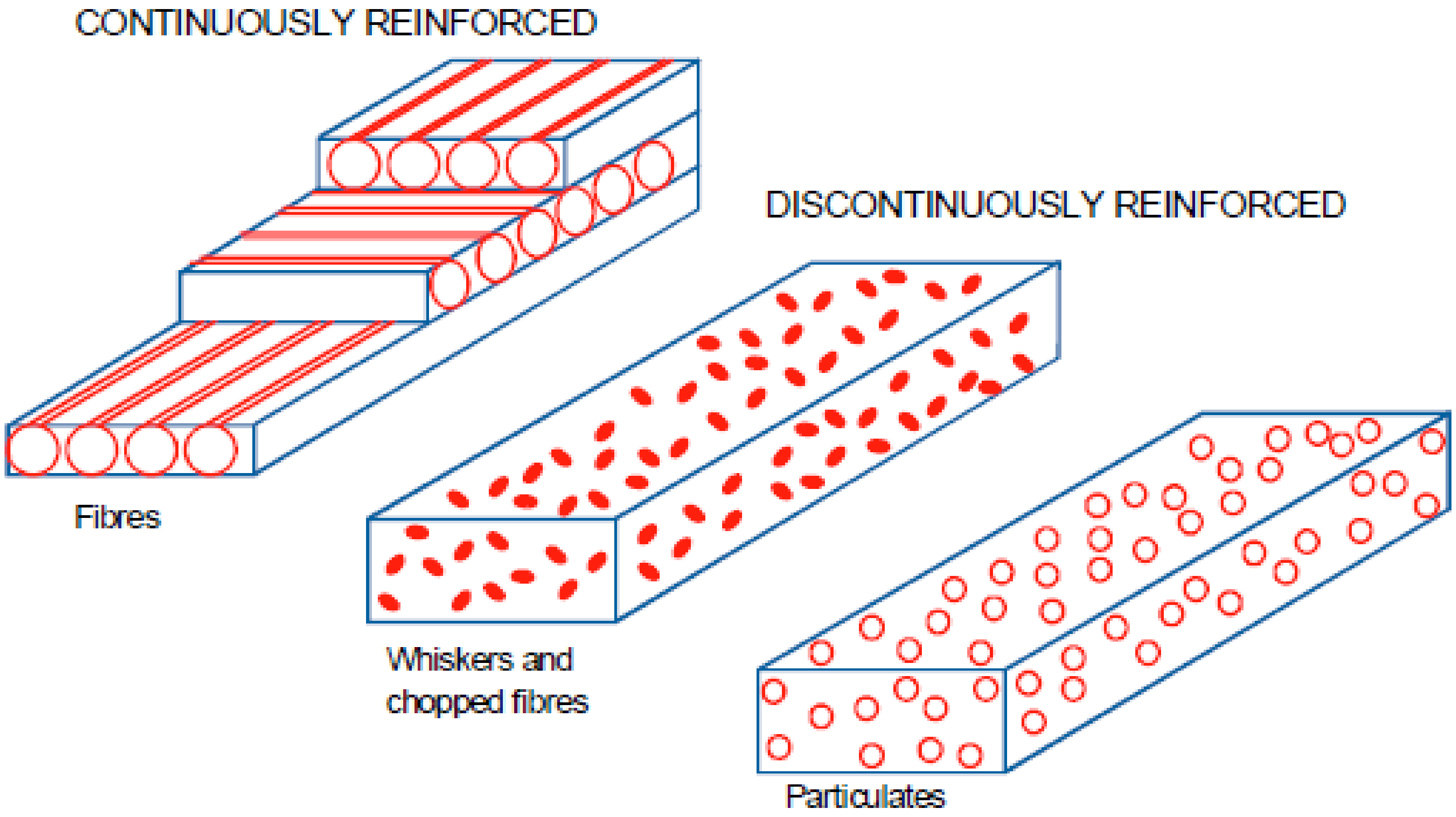
Random fiber (short fiber) reinforced composites



Continuous fiber (long fiber) reinforced composites



Typical Reinforcement Geometries for Composites



Dispersion strengthened composites

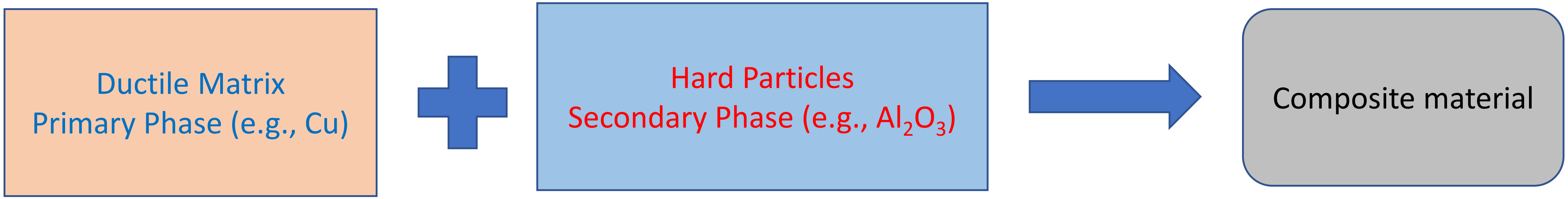
DSCs consists of a uniform dispersion of ultrafine refractory oxides in a high melting metal or alloy matrix

The particles are comparatively smaller, and are of 0.01-0.1 μm in size

The concentration of particles varies from 1 to 15 % by volume

Here the strengthening occurs at atomic/molecular level i.e., mechanism of strengthening is similar to that for precipitation hardening in metals where matrix bears the major portion of an applied load, while dispersions hinder/impede the motion of dislocation

Mechanism of dispersion strengthening



14 % Al2O3 in Al- “SAP” composite

1-2 % ThO2 in Ni-20 % Cr-TD Nickel

WC in Co-Cemented carbide cutting tools

Strengthening primarily depends on the interparticle spacing in accordance with the following relationship proposed by Orowan

$$\tau = Gb/l$$

Where G is the shear modulus of the matrix

b is the burgers vector of the dislocations

l is the interparticle spacing and τ is the shear stress required to yield and cause dislocation to move

Concept of degree of flexibility

Types of fibers

1. Glass Fibers
2. Boron Fibers
3. Carbon Fibers
4. Natural or organic fibers
5. Hybrid Fibers

Glass fibers

1. It consisting of various extremely fine fibers of glass
2. The GF can be produced by millennia technique of heat and drawing glass into fibers
3. It is most commercially viable reinforced so far
4. Recently the uses of GF also introduced in textile industries
5. It is expected that by 2022 the demand for glass fiber will jump by 4.8 % (i.e. 9.4 billon) globally

Characteristics properties of Glass fibers

They exhibit useful bulk properties

1. Hardness
2. Transparency
3. Resistance to chemical attack
4. Stability, and inertness, as well as desirable fiber properties such as strength, flexibility, and stiffness.

Glass fibers are used in the manufacture of structural composites, printed circuit boards and a wide range of special-purpose products

Over 95% of the fibers used in reinforced plastics are glass fibers, as they are inexpensive, easy to manufacture and possess high strength and stiffness with respect to the plastics with which they are reinforced

Historical significance

1. In 1893, Edward Drummond Libbey exhibited a dress at the “World Columbian exposition” incorporating glass fibers with the diameter and texture of silk fibers.

    This was first worn by the popular stage actress of the time Georgia Cayvan shown in Fig. 1.

2. The ancient Egyptians was the first civilization that made glass and turn it into glass fibres. They used it for decoration: unaware of the potential that lay within it.

3. In 1936, Owens-corning company patented the glass fiber with only $ 1 and still this produces large amount glass fiber

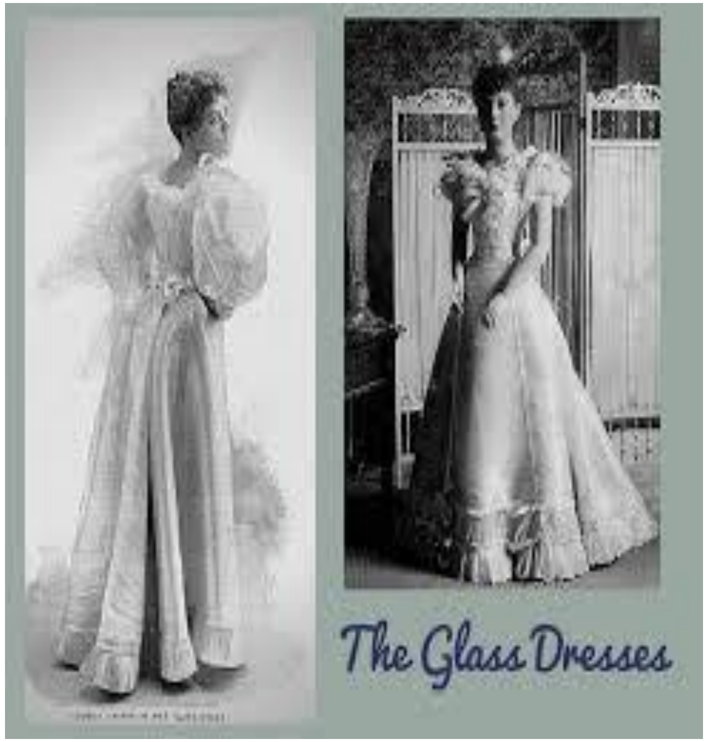
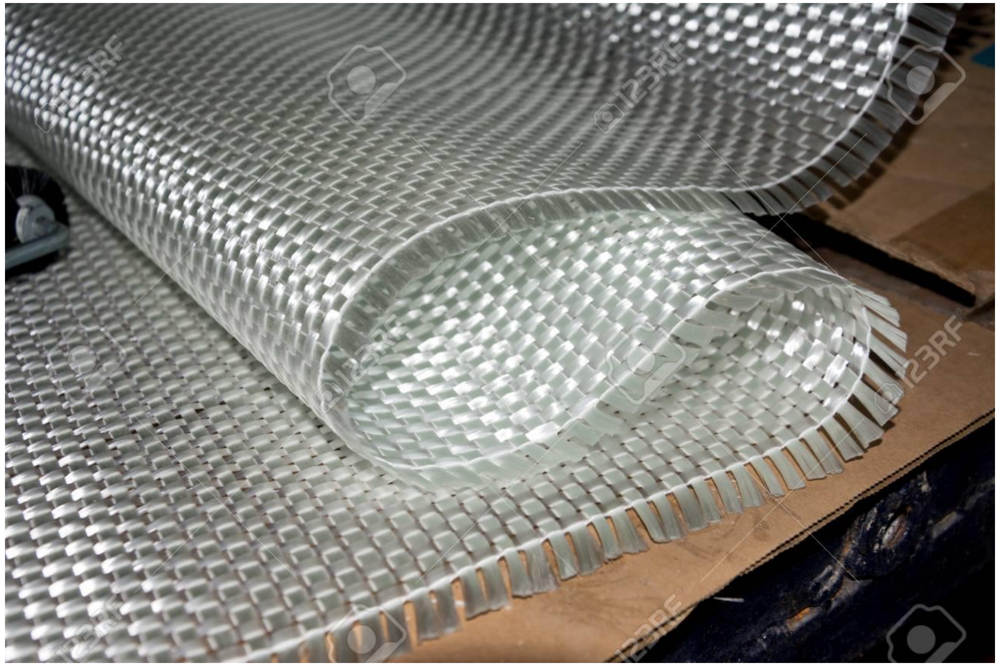


Fig. 1 The Glass dresses



(a) White glass fiber composites

(b) Glass fiber types used in structural FRP reinforcement

Glass Fiber Types

Glass fibers fall into two categories

(1) Low-cost general-purpose fibers

(2) Premium special-purpose fibers.

Over 90% of all glass fibers are general-purpose products.

These fibers are known by the designation E-glass and are subject to ASTM specifications

Many, like E-glass, have letter designations implying special properties

Letter designation	Property	Characteristic
E	Electrical	Low electrical conductivity
S	Strength	High strength
C	Chemical	High chemical durability
M	Modulus	High stiffness
A	Alkali	High alkali or soda lime glass
D	Dielectric	Low dielectric constant

Composition of glass fiber

Fiber	Ref	Composition, wt%												
		SiO_2	B_2O_3	Al_2O_3	CaO	MgO	ZnO	TiO_2	Zr_2O_3	Na_2O	K_2O	Li_2O	Fe_2O_3	F_2
General-purpose fibers														
Boron-containing E-glass	1, 2	52–56	4–6	12–15	21–23	0.4–4	...	0.2–0.5	...	0–1	Trace	...	0.2–0.4	0.2–0.7
Boron-free E-glass	7	59.0	...	12.1	22.6	3.4	...	1.5	...	0.9	...	...	0.2	...
	8	60.1	...	13.2	22.1	3.1	...	0.5	...	0.6	0.2	...	0.2	0.1
Special-purpose fibers														
ECR-glass	1, 2	58.2	...	11.6	21.7	2.0	2.9	2.5	...	1.0	0.2	...	0.1	Trace
D-glass	1, 2	74.5	22.0	0.3	0.5	...	...	...	...	1.0	<1.3	...	...	...
	2	55.7	26.5	13.7	2.8	1.0	...	...	...	0.1	0.1	0.1	...	...
S-, R-, and Te-glass	1, 2	60–65.5	...	23–25	0–9	6–11	...	...	0–1	0–0.1	...	...	0–0.1	...
Silica/quartz	1, 2	99.9999	...	...	...	...	...	...	...	...	...	...	...	...

Physical and mechanical properties of glass fibers

Fiber	Log 3 forming temperature(a)		Liquidus temperature		Softening temperature		Annealing temperature		Straining temperature		Bulk density, annealed glass, g/cm^3
	°C	°F	°C	°F	°C	°F	°C	°F	°C	°F	
General-purpose fibers											
Boron-containing E-glass	1160–1196	2120–2185	1065–1077	1950–1970	830–860	1525–1580	657	1215	616	1140	2.54–2.55
Boron-free E-glass	1260	2300	1200	2190	916	1680	736	1355	691	1275	2.62
Special-purpose fibers											
ECR-glass	1213	2215	1159	2120	880	1615	728	1342	691	1275	2.66–2.68
D-glass	. . .	. . .	. . .	. . .	770	1420	. . .	. . .	475	885	2.16
S-glass	1565	2850	1500	2730	1056	1935	. . .	. . .	760	1400	2.48–2.49
Silica/quartz	>2300	>4170	1670	3038	. . .	. . .	. . .	. . .	. . .	. . .	2.15

Fiber	Coefficient of linear expansion, 10^{-6}/°C	Specific heat, cal/g/°C	Dielectric constant at room temperature and 1 MHz	Dielectric strength, kV/cm	Volume resistivity at room temperature $\log_{10}$ (Ω cm)	Refractive index (bulk)	Weight loss in 24 h in 10% H_2SO_4, %	Tensile strength at 23 °C (73 °F)		Young's modulus		Filament elongation at break, %
								MPa	ksi	GPa	10^6 psi	
General-purpose fibers												
Boron-containing E-glass	4.9–6.0	0.192	5.86–6.6	103	22.7–28.6	1.547	~41	3100–3800	450–551	76–78	11.0–11.3	4.5–4.9
Boron-free E-glass	6.0	. . .	7.0	102	28.1	1.560	~6	3100–3800	450–551	80–81	11.6–11.7	4.6
Special-purpose fibers												
ECR-glass	5.9	. . .	. . .	. . .	. . .	1.576	5	3100–3800	450–551	80–81	11.6–11.7	4.5–4.9
D-glass	3.1	0.175	3.56–3.62	. . .	. . .	1.47	. . .	2410	349	. . .	. . .	. . .
S-glass	2.9	0.176	4.53–4.6	130	. . .	1.523	. . .	4380–4590	635–666	88–91	12.8–13.2	5.4–5.8
Silica/quartz	0.54	. . .	3.78	. . .	. . .	1.4585	. . .	3400	493	69	10.0	5

(a) The log 3 forming temperature is the temperature of a melt at a reference viscosity of 100 Pa · s (1000 P). Source: Ref 2, 7–10, 11

Fiber Forming Processes

Glass melts are made by fusing (co-melting) silica with minerals, which contain the oxides needed to form a given composition.

The molten mass is rapidly cooled to prevent crystallization and formed into glass fibers by a process also known as fiberization.

Nearly all continuous glass fibers are made by a direct draw process and formed by extruding molten glass through a platinum alloy bushing that may contain up to several thousand individual orifices, each being 0.793 to 3.175 mm (0.0312 to 0.125 in.) in diameter.

While still highly viscous, the resulting fibers are rapidly drawn to a fine diameter and solidify. Typical fiber diameters range from 3 to 20 lm (118 to 787 lin.).

Individual filaments are combined into multifilament strands, which are pulled by mechanical winders at velocities of up to 61 m/ s (200 ft/s) and wound onto tubes or forming packages.

Glass melting

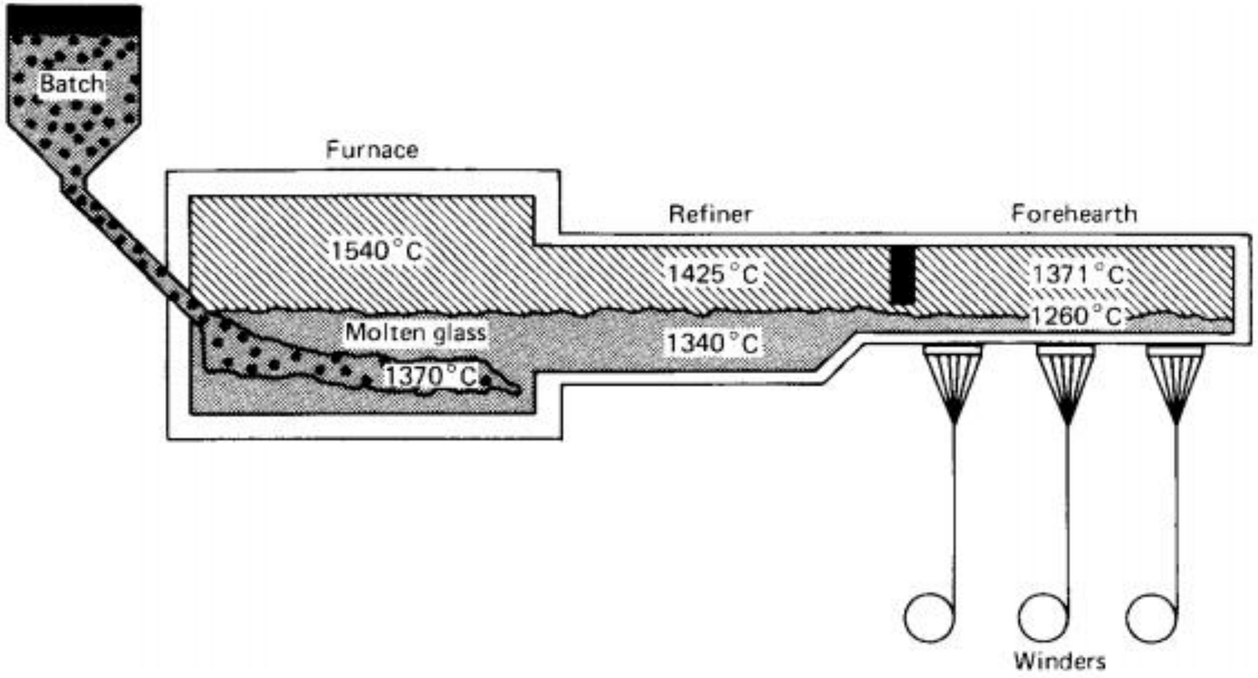


Fig. Furnace for glass melting

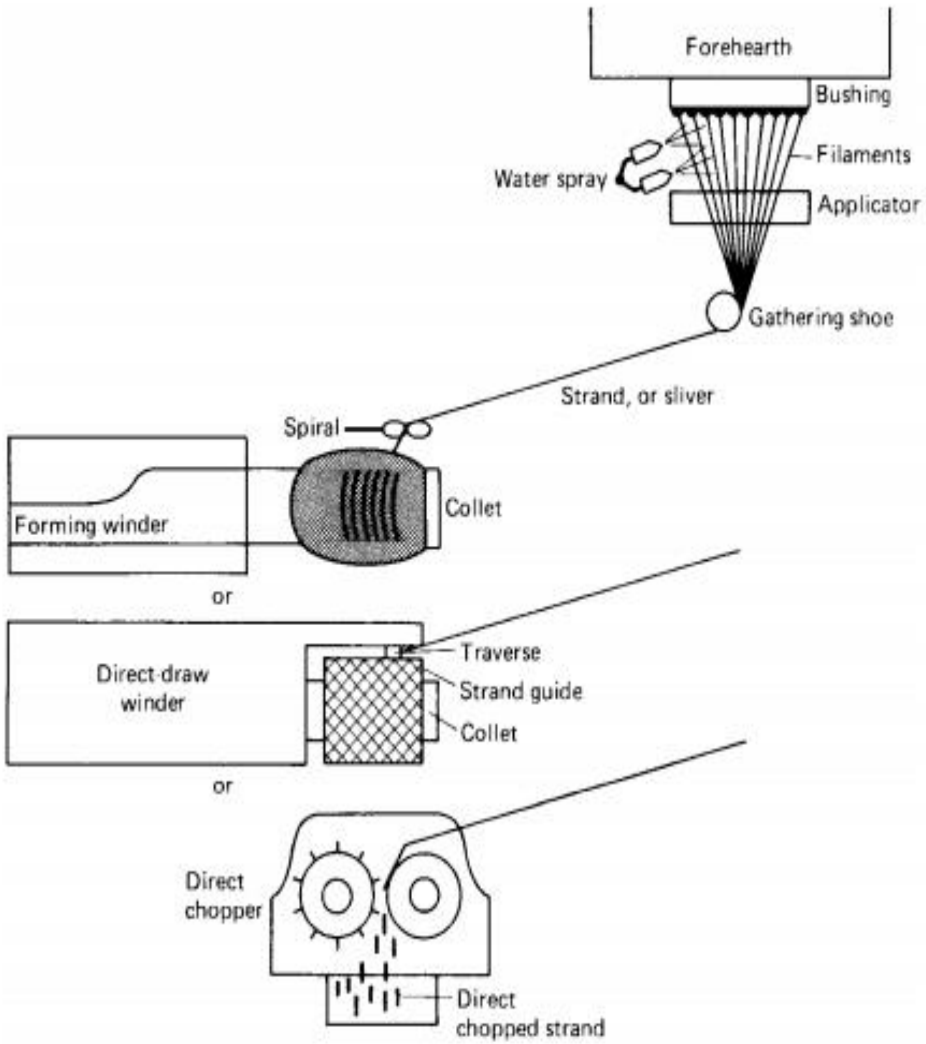
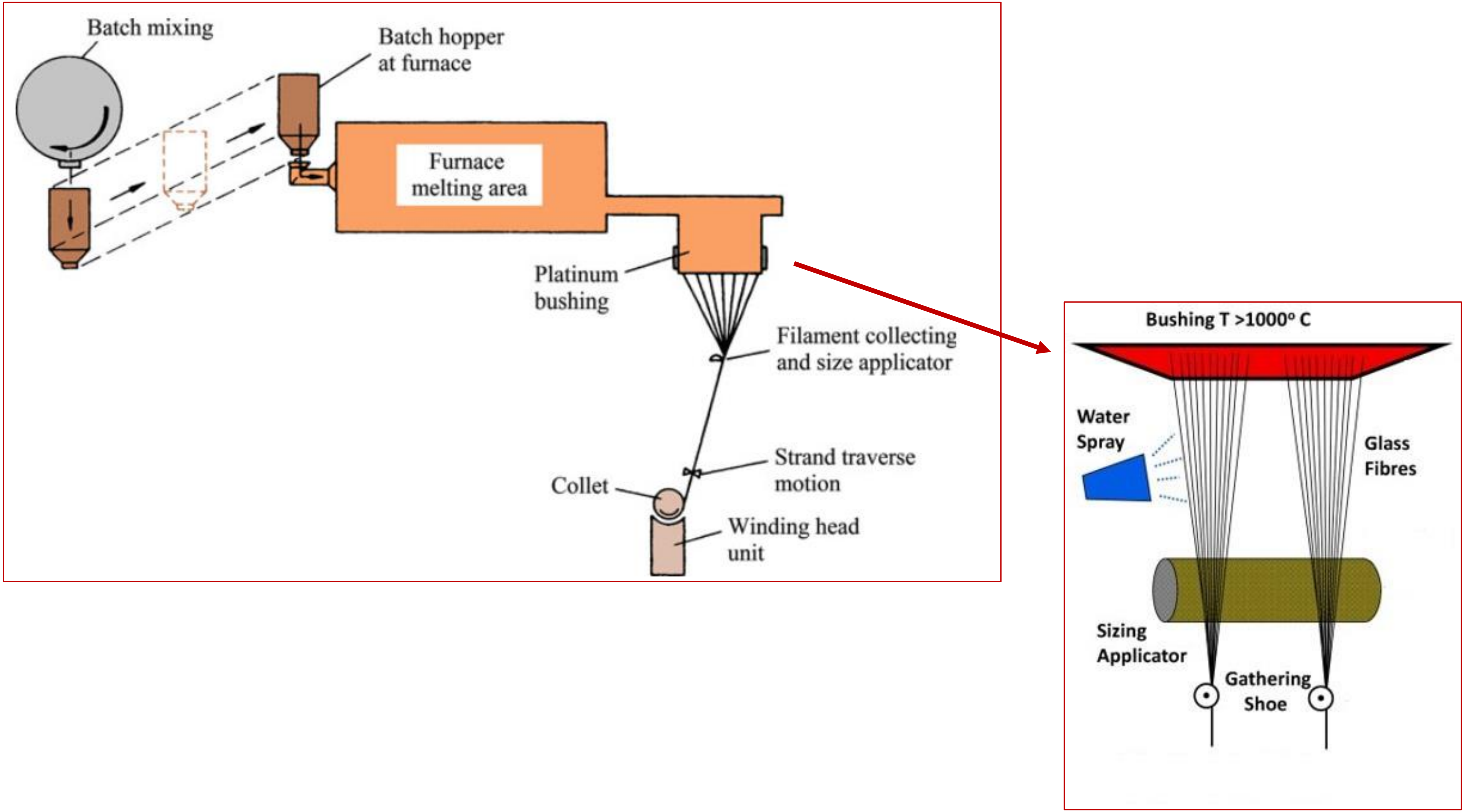
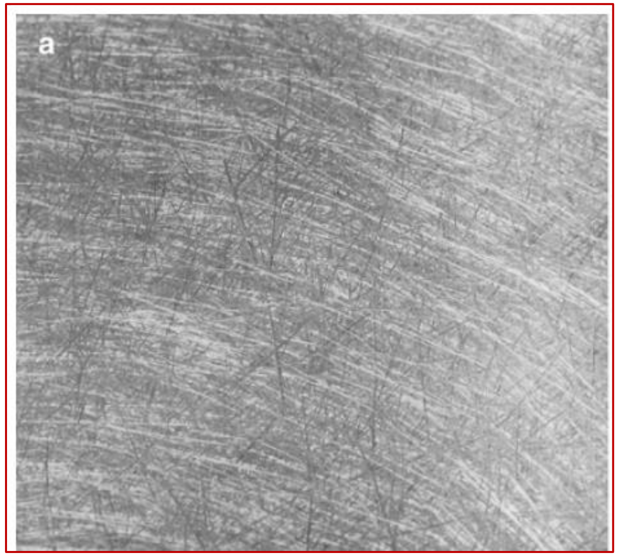


Fig. Fiber glass forming process



Batch mixing
Batch hopper at furnace
Furnace melting area
Platinum bushing
Filament collecting and size applicator
Strand traverse motion
Collet
Winding head unit
Bushing T >1000° C
Water Spray
Glass Fibres
Sizing Applicator
Gathering Shoe

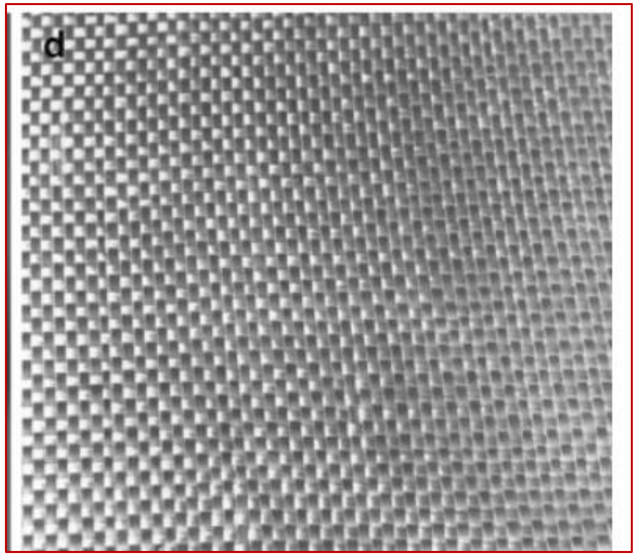
glass fiber is commercially available



(a) chopped strand

(b) continuous yarn

(c) roving



(d) fabric

- Continuous strand is a group of individual fibers;
- Roving is a group of parallel strands;
- Chopped fibers consists of strand or roving chopped to lengths between 5 and 50 mm.
- Glass fibers are also available in the form of oven fabrics or nonwoven mat

Continuous glass fibers

- A sol is a colloidal suspension in which the individual particles are so small (generally in the nm range) that they show no sedimentation.
- A gel, on the other hand, is a suspension in which the liquid medium has become viscous enough to behave more or less like a solid. The sol–gel process of making a fiber involves a conversion of fibrous gels, drawn from a solution at a low temperature, into glass or ceramic fibers at several hundred degrees Celsius.
- The maximum heating temperature in this process is much lower than that in conventional glass fiber manufacture.
- The sol–gel method using metal alkoxides consists of preparing an appropriate homogeneous solution, changing the solution to a sol, gelling the sol, and converting the gel to glass by heating.
- The sol–gel technique is a very powerful technique for making glass and ceramic fibers.

Fiberglasses (Glass fibers reinforced polymer matrix composites) are characterized by the following properties:

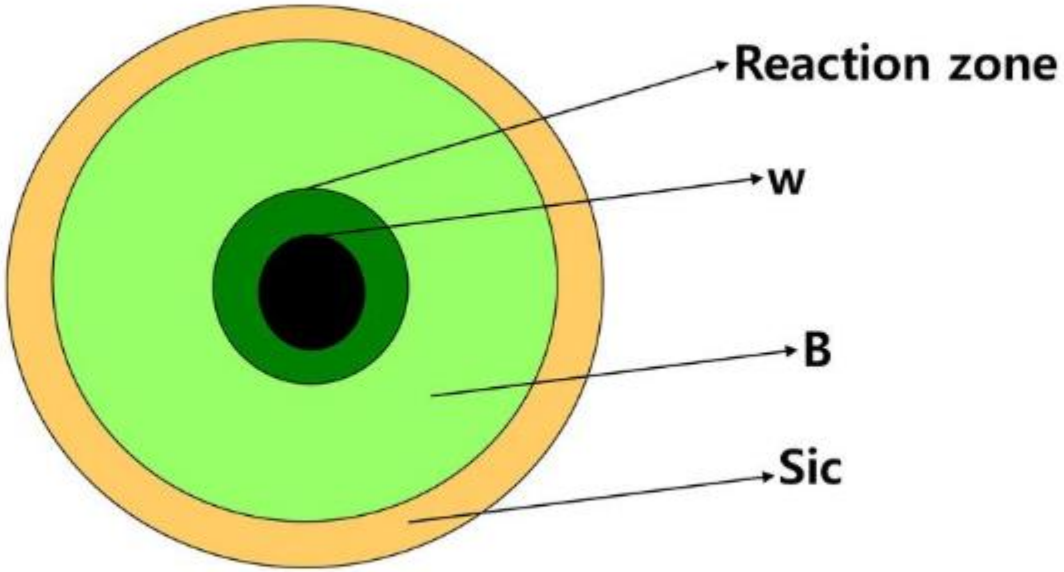
- High strength-to-weight ratio;
- High modulus of elasticity-to-weight ratio;
- Good corrosion resistance;
- Good insulating properties;
- Low thermal resistance (as compared to metals and ceramics).
- Fiberglass materials are used for manufacturing: boat hulls and marine structures, automobile and truck body panels, pressure vessels, aircraft wings and fuselage sections, housings for radar systems, swimming pools, welding helmets, roofs, pipes.

https://www.youtube.com/watch?v=wJZYtFXnnwc

Boron Fibers

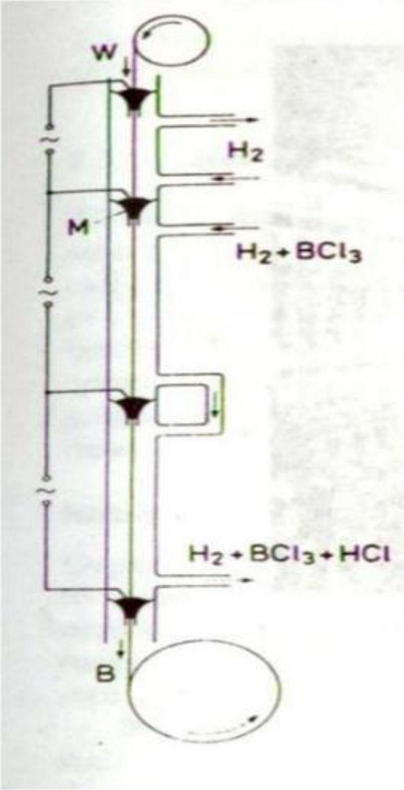
1. Five times stronger than steel
2. Twice stiffer than the steel and its alloys
3. They are made by a chemical vapor deposition process in which boron vapors are deposited onto a fine W (tungsten) or carbon filament
4. B provides strength, stiffness and light weight and possesses excellent compressive properties and buckling resistance

Fabrication of B fibers



B Fabrication

Boron Fiber Fabrication by CVD



1. Thermal Decomposition of a Boron Hydride.
Low temperatures, weak
Lack of adherence b/w **B** and **core**
Substrate : Glass fiber, low density

2. Reduction of Boron Halide.
$2BX_3 + 3H_2 = 2B + 6HX$
X : halogen elements, Cl, Br, I etc
Very high Temp. Strong, Uniform Quality
Substrate: W wire, high density

→ ***Fig. Schematic Diagram of Reduction of Boron Halide***

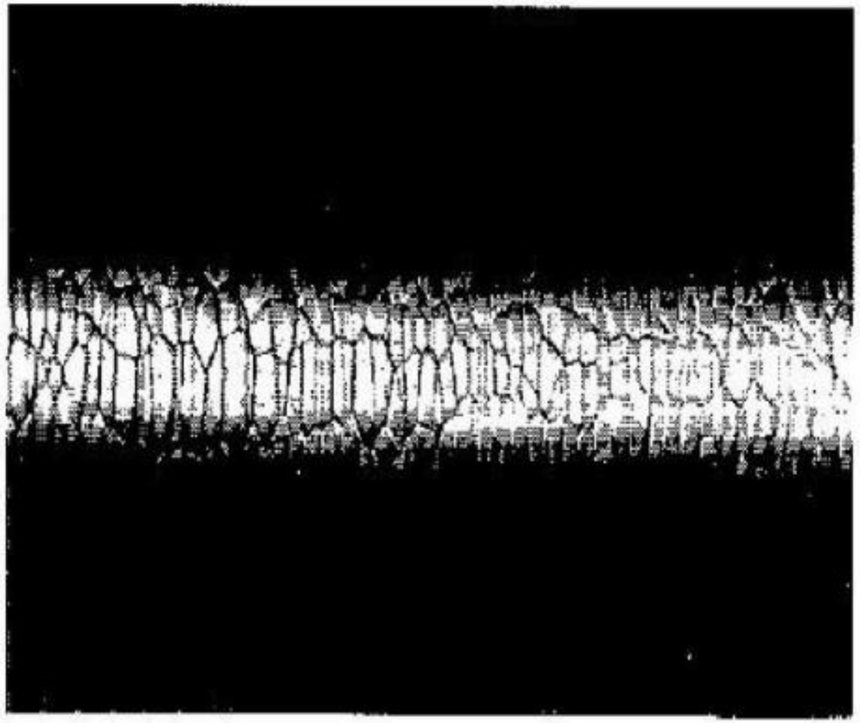


Fig. 2.11. Characteristic corn-cob structure of boron fiber. [From van Maaren et al. (1975), used with permission.]

Krishan K. Chawla, Composite materials science and engineering, Springer-Verlag, (1998)

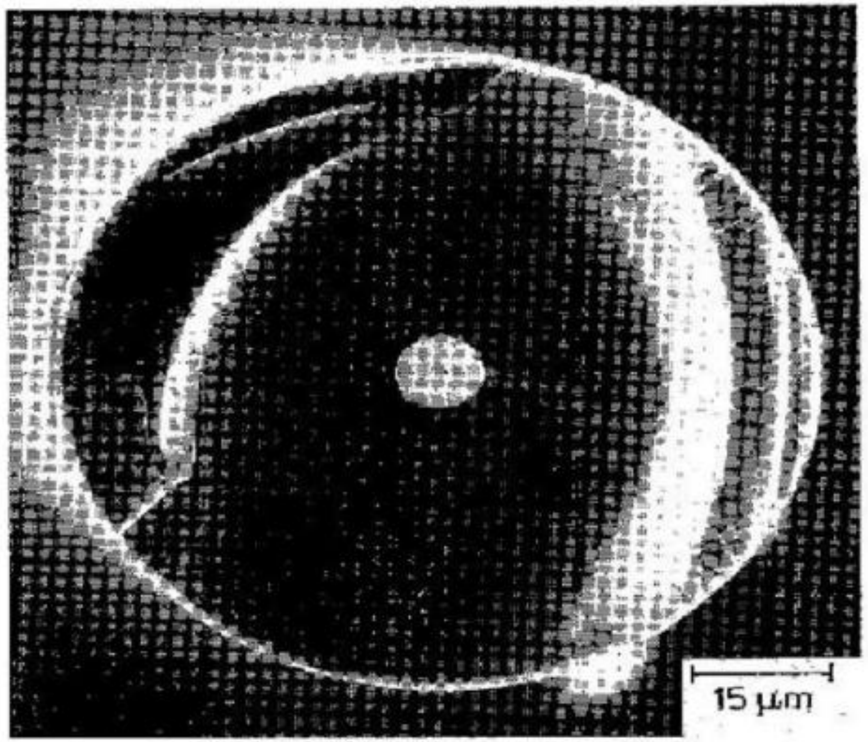


Fig. 2.13. Fracture surface of a boron fiber showing a characteristically brittle fracture and a radial crack

Strength properties of improved large dia B fibers

Diameter (μm)	Treatment	Average[a] (GPa)	COV[b] (%)	Relative fracture energy
142	As-produced	3.8	10	1.0
406	As-produced	2.1	14	0.3
382	Chemical polish	4.6	4	1.4
382	Heat treatment plus polish	5.7	4	2.2

[a] Gauge length = 25 mm.
[b] Coefficient of variation = standard deviation/average value.
Source: Reprinted with permission from *Journal of Metals*, **37**, No. 6, 1985, a publication of The Metallurgical Society, Warrendale, PA.

Ref .: K K Chwala, Composite material Science and Engineering, springer-veriag (1998)

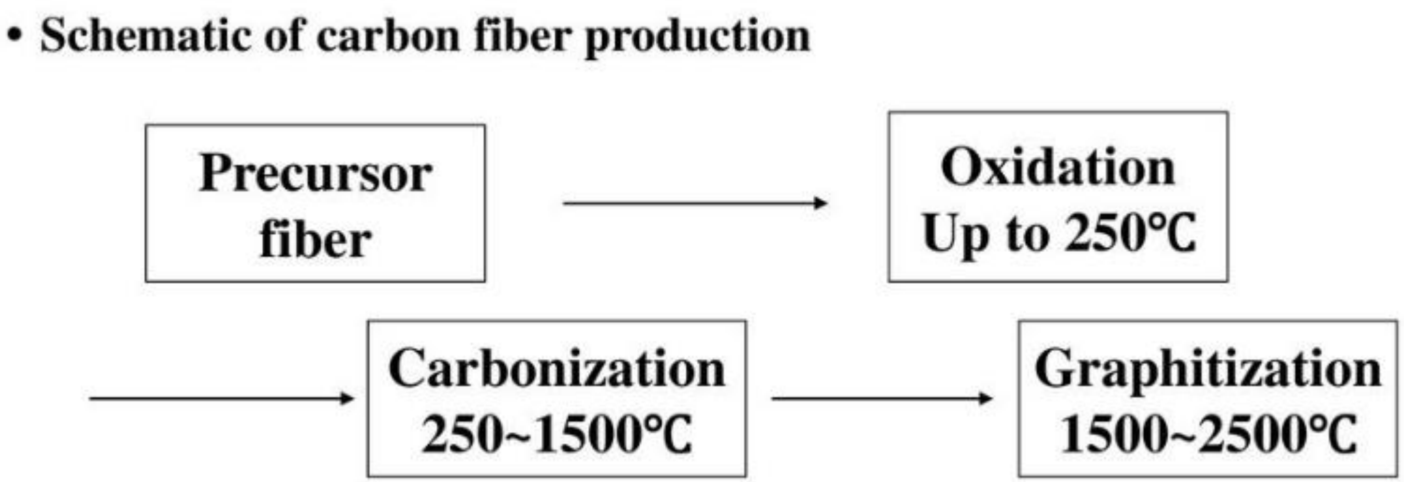
Applications of B fibers

Uses for boron composites range from sporting goods to missiles

1. fishing rod
2. golf club shafts
3. bicycle frames
4. Military aircraft
5. Tennis racket etc

Carbon fibers

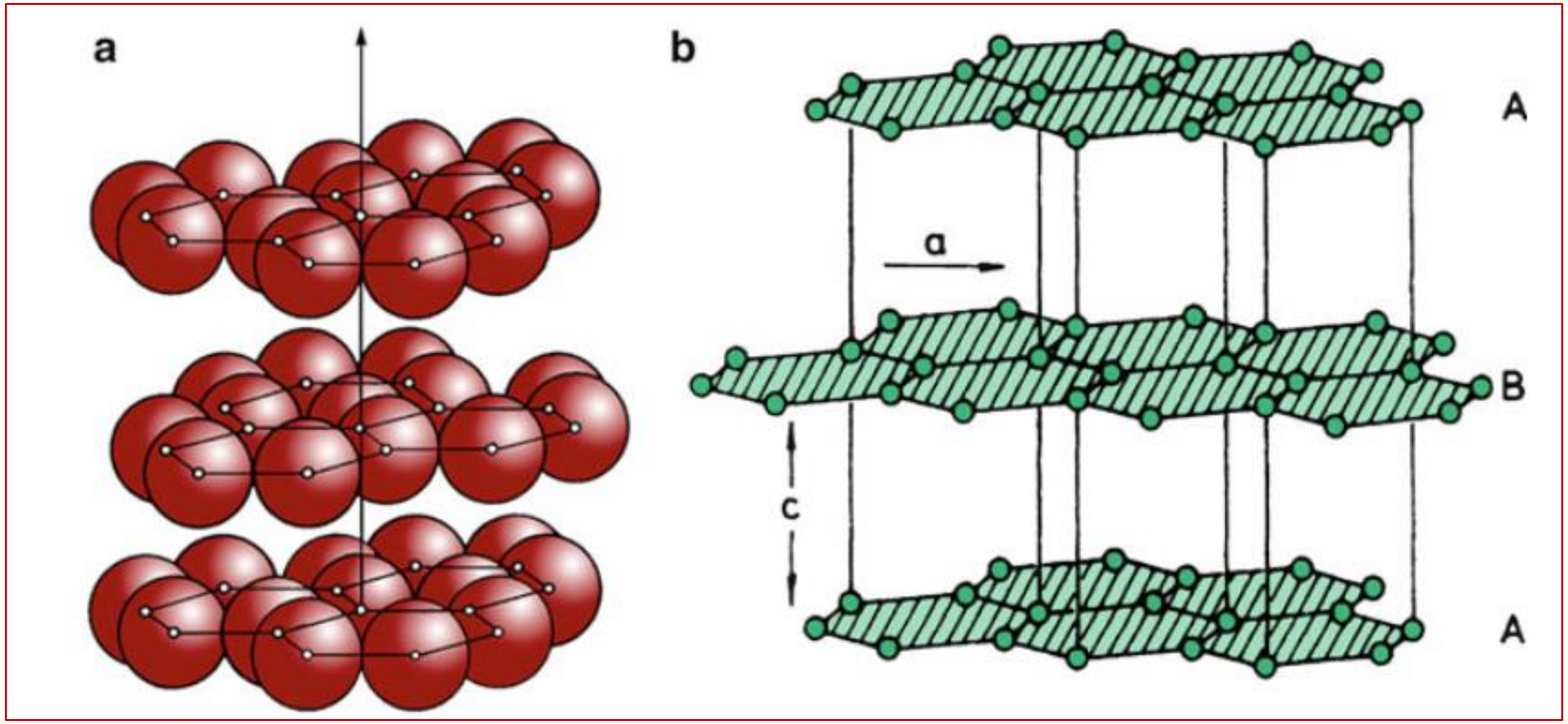
Carbon fiber is manufactured from organic precursors at high temperature with taking advantage of alignment of an axis through stabilization, carbonization and graphitization



Ref .: K K Chwala, Composite material Science and Engineering, springer-veriag (1998)

Carbon Fibers

- density equal to 2.268 g/cm^3
- highly anisotropic



(a) Graphitic layer structure. (b) hexagonal lattice structure of graphite

Most of the carbon fiber fabrication processes involve the following essential steps:

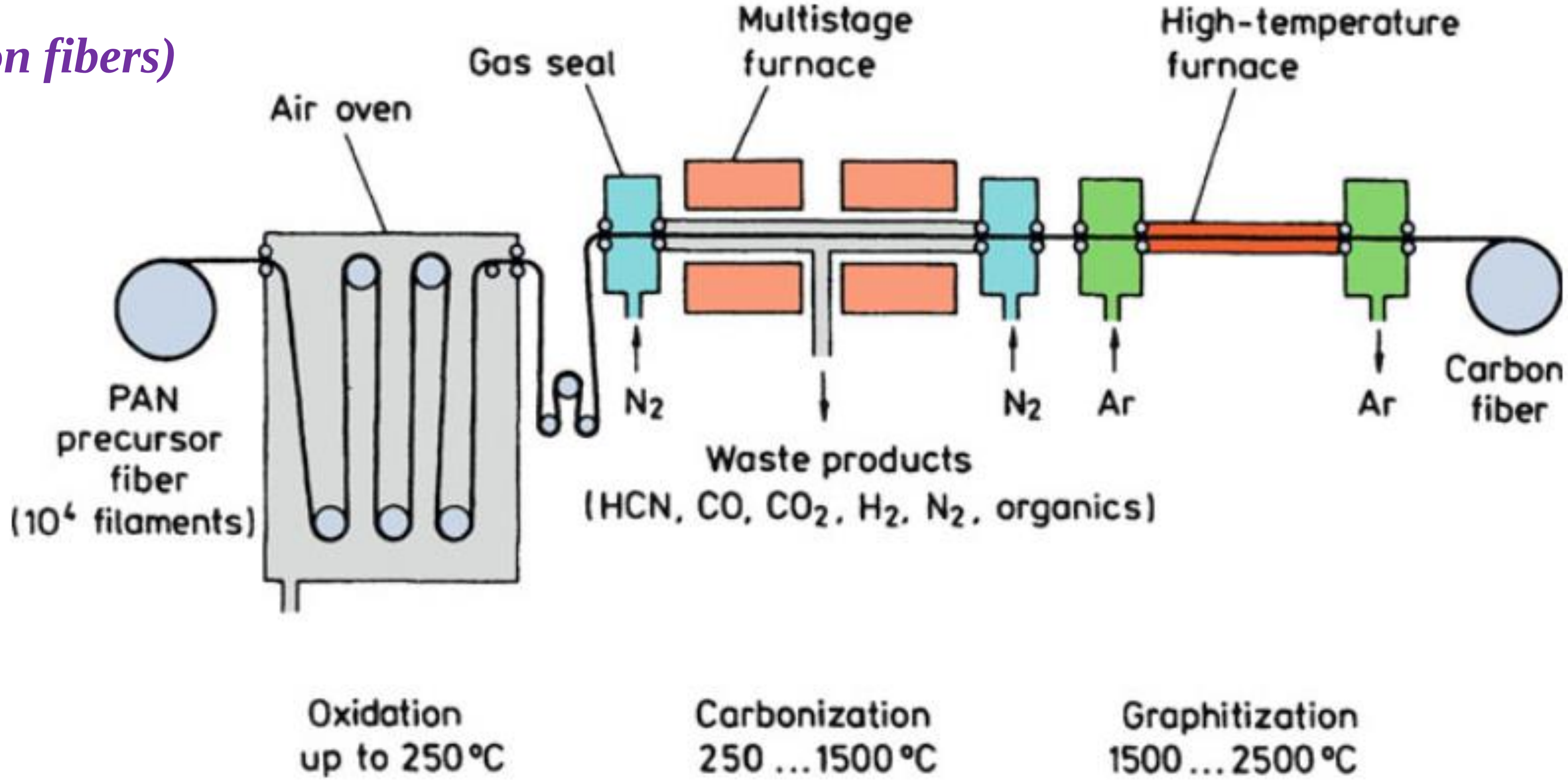
1. A fiberization procedure to make a precursor fiber. This generally involves wet-,dry-, or melt-spinning followed by some drawing or stretching.
2. A stabilization treatment that prevents the fiber from melting in the subsequent high-temperature treatments.
3. A thermal treatment called carbonization that removes most noncarbon elements.
4. An optional thermal treatment called graphitization that improves the properties of carbon fiber obtained in step 3.

Carbon fibers are also classified according to the manufacturing method:

- The raw material used to make carbon fiber is called the precursor.
- About 90% of the carbon fibers produced are made from polyacrylonitrile.
- The remaining 10% are made from rayon or petroleum pitch

1. PAN-based carbon fibers

(the most popular type of carbon fibers)



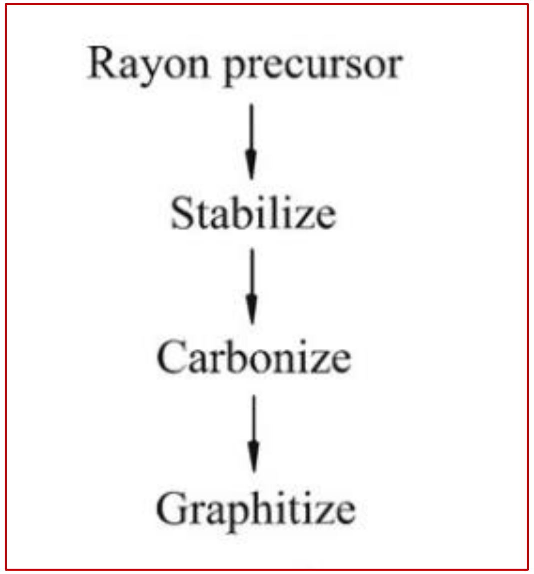
In this method carbon fibers are produced by conversion of polyacrylonitrile (PAN) precursor through the following stages:

- Stretching filaments from polyacrylonitrile precursor and their thermal oxidation at 400°F (200°C). The filaments are held in tension.
- Carbonization in Nitrogen atmosphere at a temperature about 2200 °F (1200°C) for several hours. During this stage non-carbon elements (O,N,H) volatilize resulting in enrichment of the fibers with carbon.
- Graphitization at about 4500 °F (2500°C).

2. Rayon-based carbon fiber production

Rayon is a thermosetting polymer

- fiberization,
- stabilization in a reactive atmosphere (air or oxygen, <400 C),
- carbonization (<1,500 C), and
- graphitization (>2,500 C).



Schematic of rayon-based carbon fiber production

Carbon Fiber Reinforced Polymers (CFRP) are characterized by the following properties:

- Light weight; High strength-to-weight ratio;
- Very High modulus elasticity-to-weight ratio;
- High Fatigue strength; Good corrosion resistance;
- Very low coefficient of thermal expansion;
- Low impact resistance; High electric conductivity;
- High cost.
- Carbon Fiber Reinforced Polymers (CFRP) are used for manufacturing: automotive marine and aerospace parts, sport goods (golf clubs, skis, tennis racquets, fishing rods), bicycle frames.

- *The types of carbon fibers are as follows:*
- **UHM** (ultra high modulus). Modulus of elasticity > 65400 ksi (450GPa).
- **HM** (high modulus). Modulus of elasticity is in the range 51000-65400 ksi (350-450GPa).
- **IM** (intermediate modulus). Modulus of elasticity is in the range 29000-51000 ksi (200-350GPa).
- **HT** (high tensile, low modulus). Tensile strength > 436 ksi (3 GPa), modulus of elasticity < 14500 ksi (100 GPa).
- **SHT** (super high tensile). Tensile strength > 650 ksi (4.5GPa).

Carbon/Graphite Fibers

Manufacturing process of carbon/graphite fibers.

1. Polyacrylonitrile (PAN): good properties with relatively low cost

2. Rayon (regenerated cellulose): high and uniform properties but more expensive than Pitch

3. Pitch: lower in cost

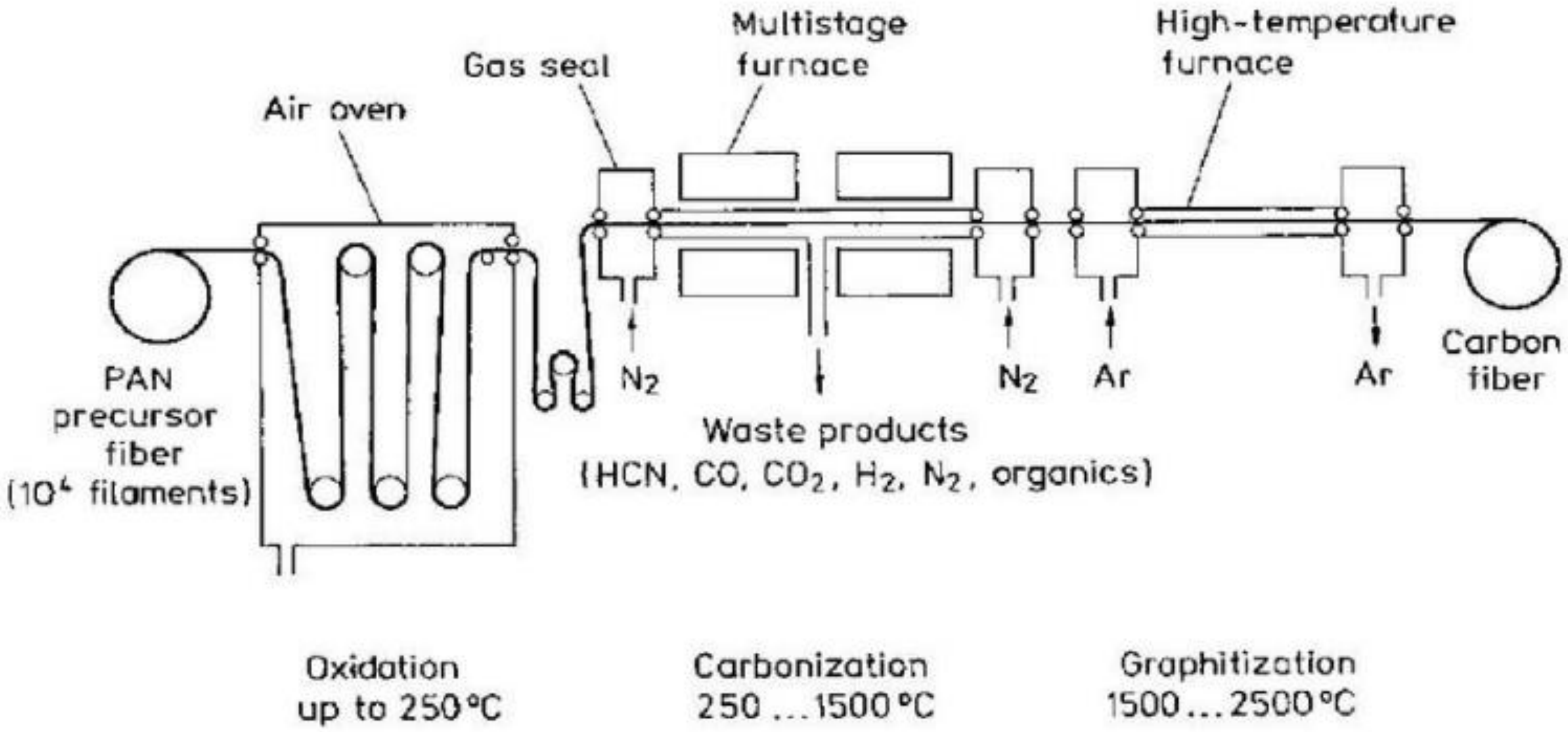


Fig. 2.15. Schematic of PAN-based carbon fiber production. [Reprinted with permission from Baker (1983).]

Fig. Schematic of PAN based carbon fiber production

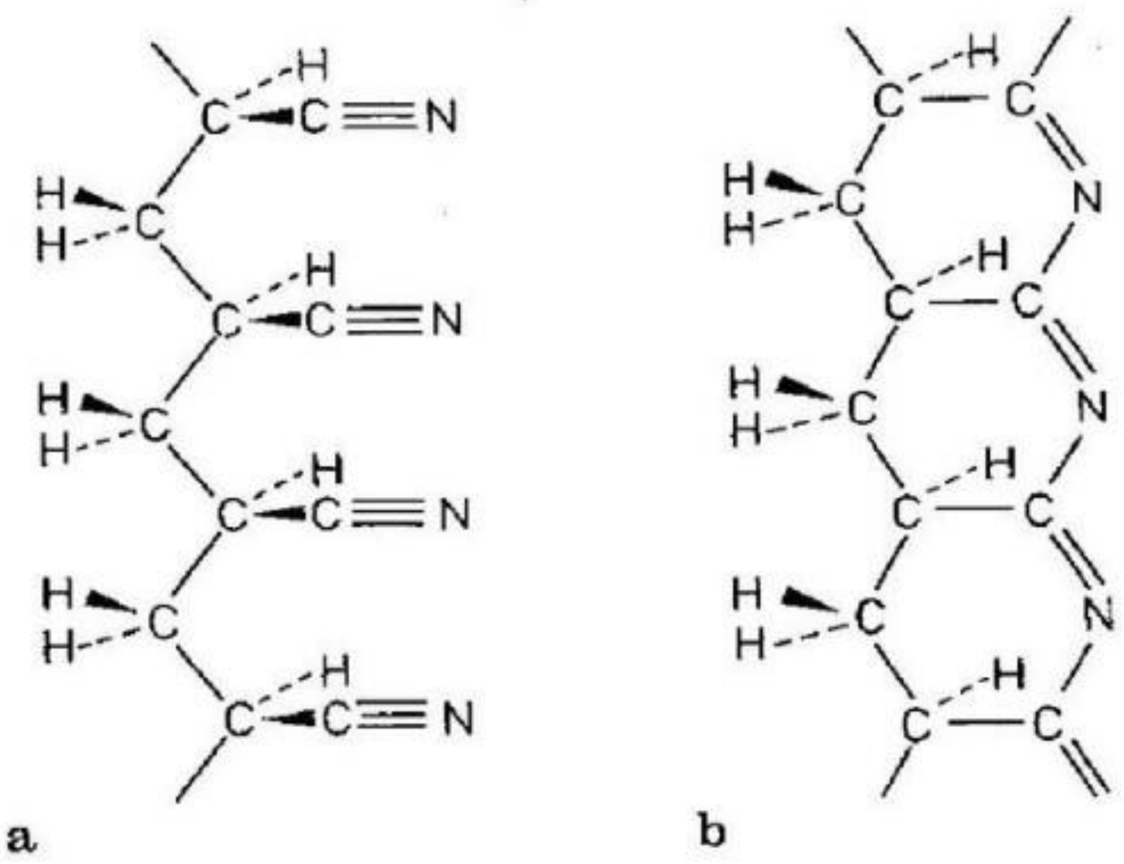


Fig. 2.16 a flexible polyacrylonitrile molecule b. Rigid ladder (or oriented cyclic)

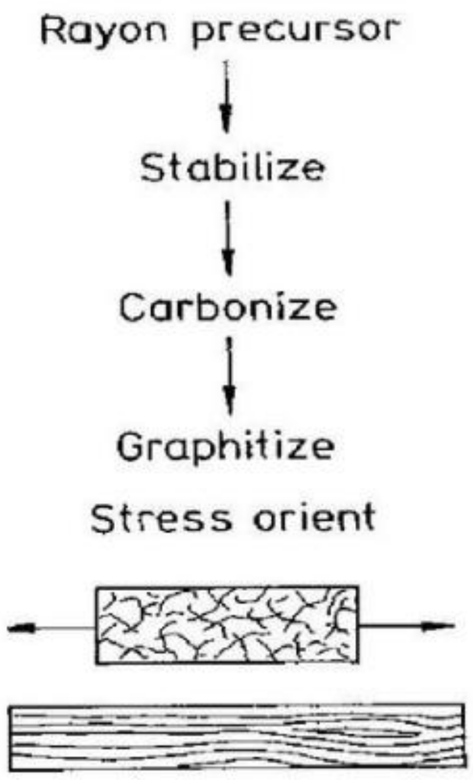


Fig. 2.18. Schematic of rayon-based carbon fiber production. [After Diefendorf and Tokarsky (1975), used with permission.]

Ref .: K K Chwala, Composite material Science and Engineering, springer-veriag (1998)

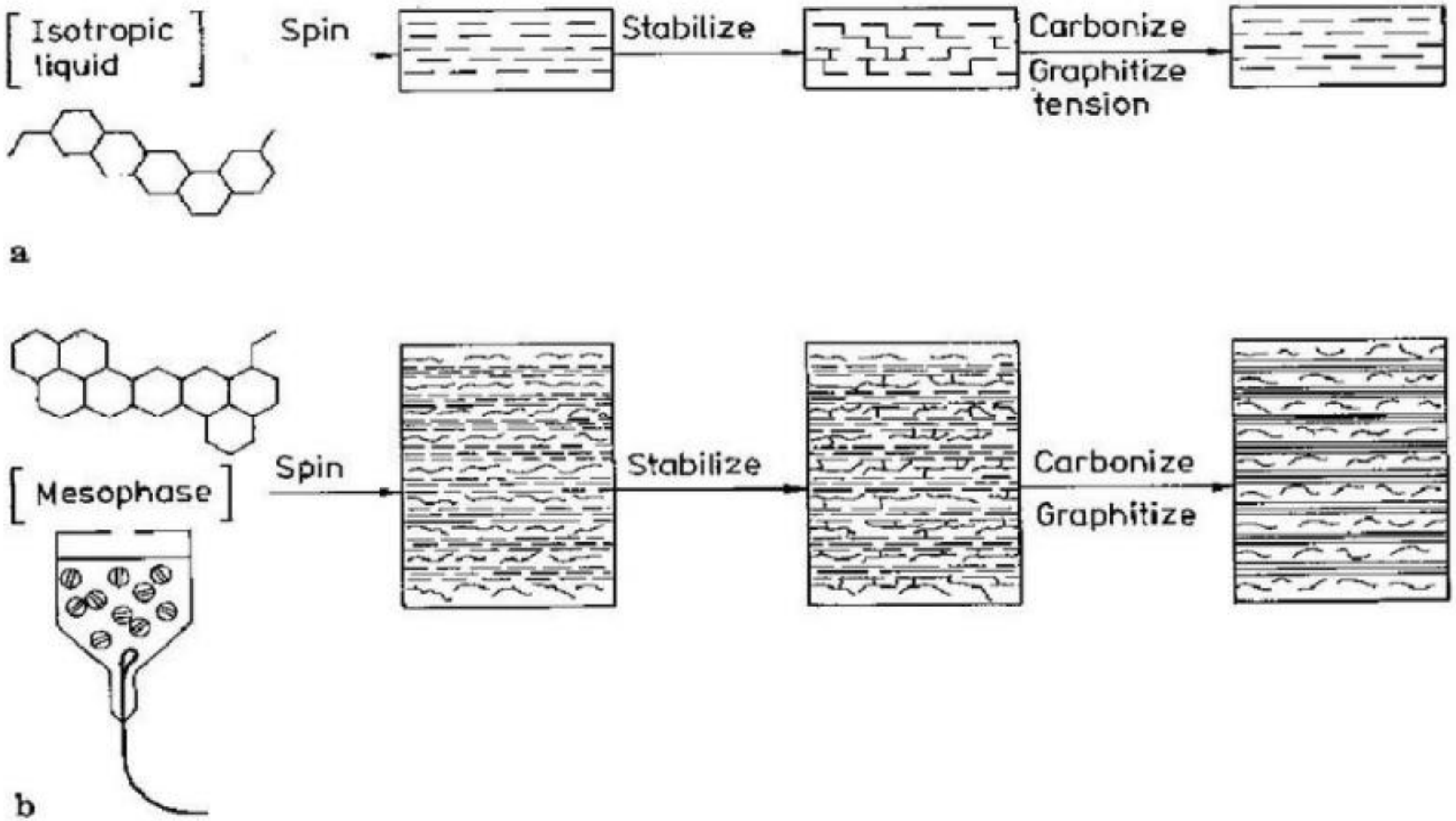


Fig. 2.19. Schematic of pitch-based carbon fiber production: **a** isotropic pitch process, **b** mesophase pitch process. [With permission from Diefendorf and Tokarsky (1975).]

Carbon Fiber Mechanical Properties

PAN-BASED FIBERS	LOW MODULUS	HIGH MODULUS
Tensile Modulus (ksi)	33	56
Tensile Strength (ksi)	0.48	0.35
Elongation (%)	1.4	0.6
Density (g/cc)	1.8	1.9
Carbon Assay (%)	92-97	100
PITCH-BASED FIBERS		
Tensile Modulus (ksi)	23	55
Tensile Strength (ksi)	0.2	0.25
Elongation (%)	0.9	0.4
Density (g/cc)	1.9	2.0
Carbon Assay (%)	97	99
RAYON-BASED FIBERS		
Tensile Modulus (ksi)	5.9	—
Tensile Strength (ksi)	0.15	—
Elongation (%)	2.5	—
Density (g/cc)	1.6	—
Carbon Assay (%)	99	—

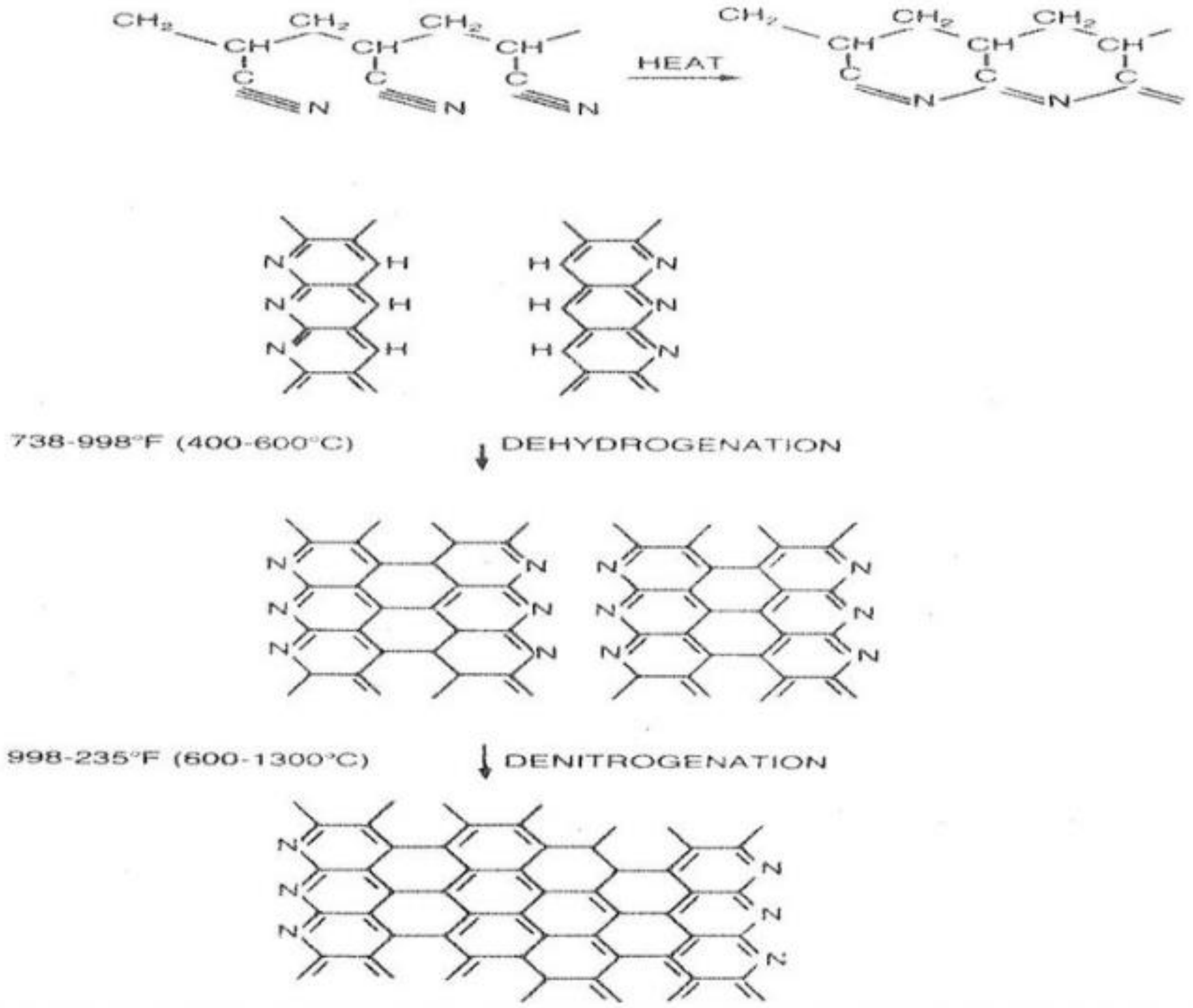
Comparison of properties of different carbon fibers

Table 2.4. Comparison of properties of different carbon fibers

Precursor	Density ($g\ cm^{-3}$)	Young's modulus (GPa)	Electrical resistivity ($10^{-4}\ \Omega\ cm$)
Rayon[a]	1.66	390	10
Polyacrylonitrile[b] (PAN)	1.74	230	18
Pitch (Kureha)			
LT[c]	1.6	41	100
HT[d]	1.6	41	50
Mesophase pitch[e]			
LT	2.1	340	9
HT	2.2	690	1.8
Single-crystal[f] graphite	2.25	1000	0.40

[a] Union Carbide, Thornel 50.
[b] Union Carbide, Thornel 300.
[c] LT, low-temperature heat-treated.
[d] HT, high-temperature heat-treated.
[e] Union Carbide type P fibers.
[f] Modulus and resistivity are in-plane values.
Source: Adapted with permission from Ref. 38.

Schematic Representation of Carbon Fibers Preparation from PAN



Carbon Fibers

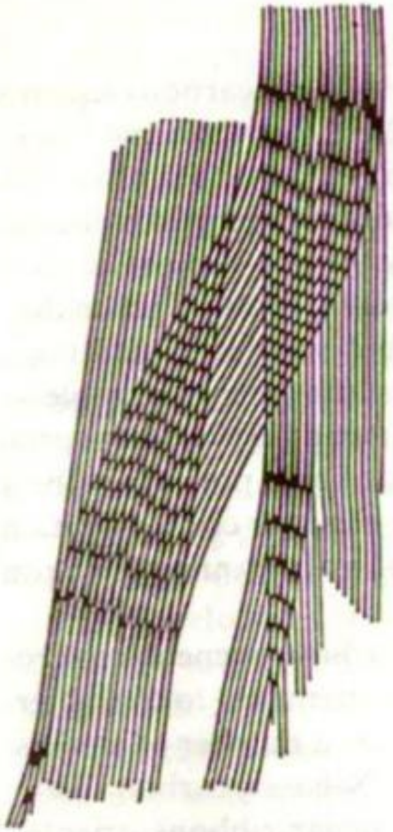


Fig. 2.21

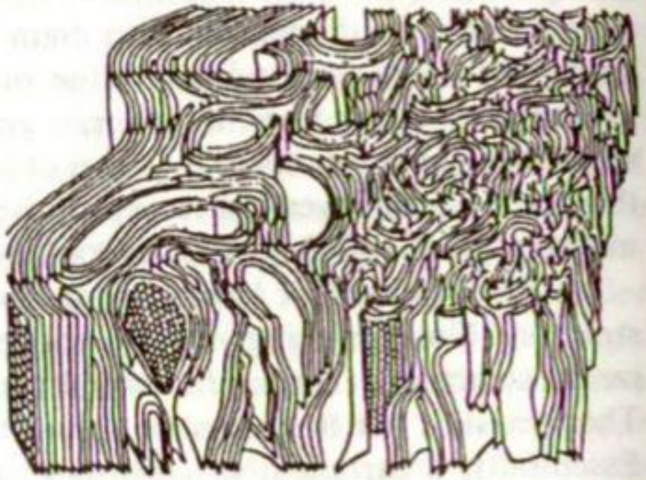


Fig. 2.22

Fig. 2.21. Two-dimensional representation of PAN-based carbon fiber. (Reprinted with permission from [Carbon, 17, S.C. Bennett and D.J. Johnson, 25], Copyright [1979], Pergamon Press, Ltd.)

Fig. 2.22. Three-dimensional representation of PAN-based carbon fiber. (From Ref. 33, used with permission.)

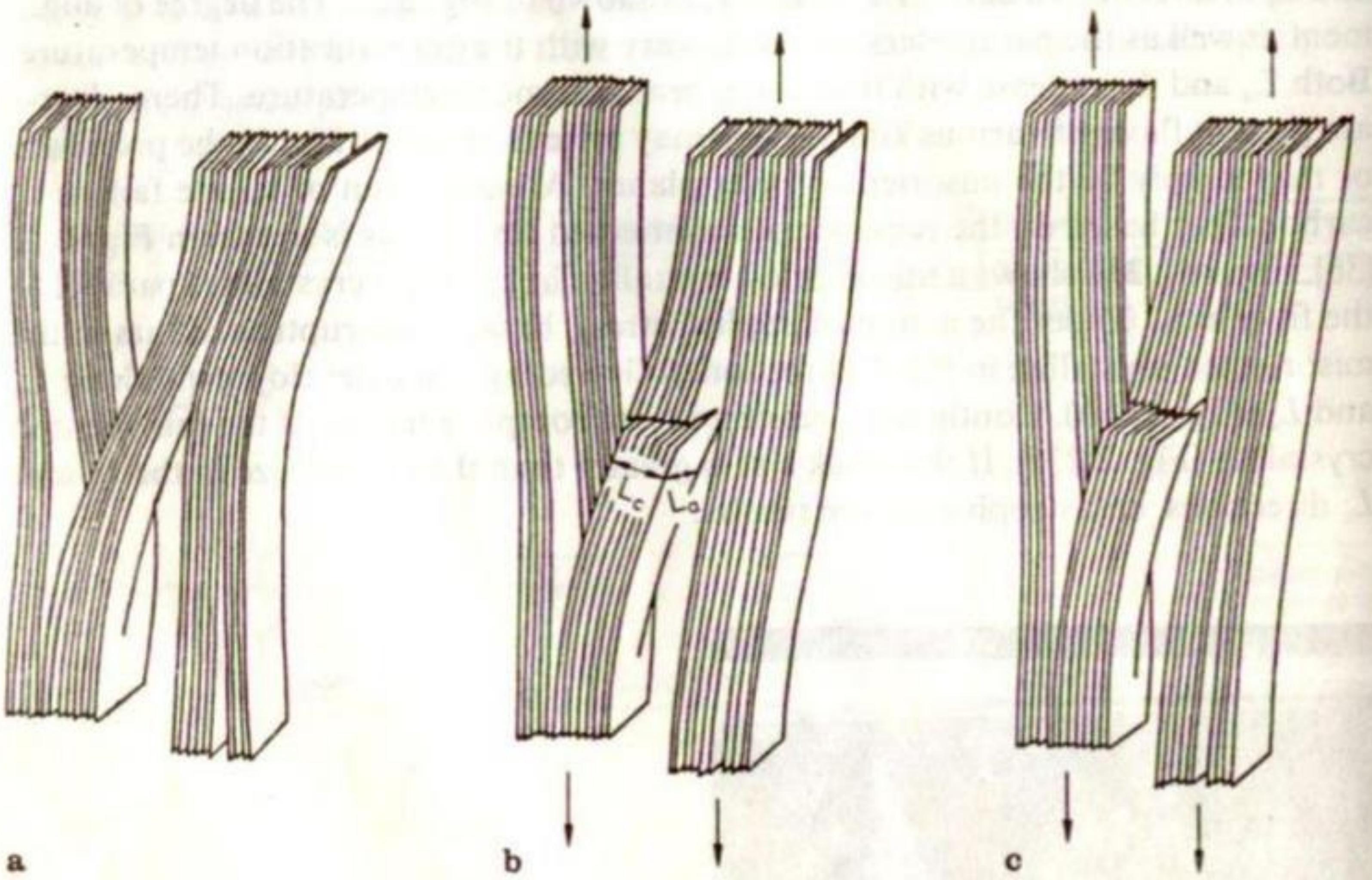


Fig. 2.23. Model for tensile failure of carbon fiber: **a** a misoriented crystallite linking two crystallites parallel to the fiber axis, **b** basal plane rupture under the action of applied stress, **c** complete failure of the misoriented crystallite. (From Ref. 36, used with permission.)

Kevlar fibers

Kevlar - aramid fibers.

- Kevlar fibers were originally developed as a replacement of steel in automotive tires.
- Kevlar may protect carbon fibers and improve their properties: hybrid fabric (Kevlar + Carbon fibers) combines very high tensile strength with high impact and abrasion resistance.
- High tensile strength (five times stronger per weight unite than steel);
- High modulus of elasticity; Very low elongation up to breaking point; Low weight; High chemical inertness;
- Very low coefficient of thermal expansion; High Fracture Toughness (impact resistance);
- High cut resistance; Textile processibility; Flame resistance.
- The disadvantages of Kevlar are: ability to absorb moisture (making Kevlar composites more sensitive to the environment), difficulties in cutting, low compressive strength.

Table 2.6 Properties of Kevlar aramid fiber yarns[a]

Property	K 29	K 49	K 119	K 129	K 149
Density (g/cm^3)	1.44	1.45	1.44	1.45	1.47
Diameter (μm)	12	12	12	12	12
Tensile strength (GPa)	2.8	2.8	3.0	3.4	2.4
Tensile strain to fracture (%)	3.5–4.0	2.8	4.4	3.3	1.5–1.9
Tensile modulus (GPa)	65	125	55	100	147
Moisture regain (%) at 25 °C, 65 % RH	6	4.3	–	–	1.5
Coefficient of expansion (10^{-6} K^{-1})	−4.0	−4.9	–	–	–

[a]All data from Du Pont brochures. Indicative values only. 25-cm yarn length was used in tensile tests (ASTM D-885). *K* stands for Kevlar, a trademark of Du Pont

Kevlar. This is meant mainly for use as rubber reinforcement for tires (belts or radial tires for cars and carcasses of radial tires for trucks)

Kevlar 29. This is used for ropes, cables, coated fabrics for inflatables, architectural fabrics, and ballistic protection fabrics. Vests made of Kevlar 29 have been used by law-enforcement agencies in many countries.

Kevlar 49. This is meant for reinforcement of epoxy, polyester, and other resins for use in the aerospace, marine, automotive, and sports industries.

Applications of B fibers

Uses for boron composites range from sporting goods to missiles

1. fishing rod
2. golf club shafts
3. bicycle frames
4. Military aircraft
5. Tennis racket etc

Properties of various fibers

Material	Tensile Modulus (E) (GPa)	Tensile Strength (σ_u) (GPa)	Density (ρ) (g/cm^3)	Specific Modulus (E/ρ)	Specific Strength (σ_u/ρ)
Fibers					
E-glass	72.4	3.5[a]	2.54	28.5	1.38
S-glass	85.5	4.6[a]	2.48	34.5	1.85
Graphite (high modulus)	390.0	2.1	1.90	205.0	1.1
Graphite (high tensile strength)	240.0	2.5	1.90	126.0	1.3
Boron	385.0	2.8	2.63	146.0	1.1
Silica	72.4	5.8	2.19	33.0	2.65
Tungsten	414.0	4.2	19.30	21.0	0.22
Beryllium	240.0	1.3	1.83	131.0	0.71
Kevlar 49 (aramid polymer)	130.0	2.8	1.50	87.0	1.87
Conventional materials					
Steel	210.0	0.34–2.1	7.8	26.9	0.043–0.2[illegible]
Aluminum alloys	70.0	0.14–0.62	2.7	25.9	0.052–0.2[illegible]


THANK YOU